

CFD 2025-26 – Lab 05

Penalty method for Dirichlet BCs

Consider the problem

$$\begin{cases} -\Delta \mathbf{u} + \nabla p = \mathbf{0} & \text{in } \Omega, \\ \operatorname{div} \mathbf{u} = 0 & \text{in } \Omega, \\ \mathbf{u} = \mathbf{g} & \text{on } \Gamma_D, \\ \mathbf{n} \cdot \nabla \mathbf{u} - p \mathbf{n} = \mathbf{h} & \text{on } \Gamma_N = \partial\Omega \setminus \Gamma_D, \end{cases}$$

with given boundary data functions $\mathbf{g}, \mathbf{h}$. As in previous labs, to impose Dirichlet BCs *in a strong sense*, we should have $V_0 = [H_{\Gamma_D}^1(\Omega)]^2$ as test space. Using the penalty method, instead, we impose also Dirichlet boundary conditions *weakly*. Perturbation of the Dirichlet condition by a small $\varepsilon > 0$:

$$\mathbf{u} = \mathbf{g} + \varepsilon(\nabla \mathbf{u} - p \mathbf{I}) \mathbf{n} \quad \text{on } \Gamma_D \quad \longrightarrow \quad (\nabla \mathbf{u} - p \mathbf{I}) \mathbf{n} = \frac{1}{\varepsilon}(\mathbf{u} - \mathbf{g}) \quad \text{on } \Gamma_D \quad (1)$$

Remark: In the actual implementation of this method, we are going to use an extremely small value $\varepsilon = 10^{-20}$. In this way, the perturbation is below the order of machine precision, but the imposition of the conditions is extremely effective. Indeed, denoting by $\bar{i}$ an index corresponding to a velocity d.o.f. lying on the Dirichlet boundary, the $\bar{i}$ -th row of the algebraic formulation of the FEM for the problem is dominated by the terms multiplied by the very large number $\frac{1}{\varepsilon}$, thus

$$\int_{\Gamma_D} \mathbf{u} \cdot \boldsymbol{\varphi}_{\bar{i}} \simeq \int_{\Gamma_D} \mathbf{g} \cdot \boldsymbol{\varphi}_{\bar{i}}.$$

Exercise 1

Let us consider the Stokes cavity problem defined in the unit square $\Omega := (0, 1)^2$:

$$\begin{cases} -\Delta \mathbf{u} + \nabla p = 0, & \text{in } \Omega, \\ \operatorname{div} \mathbf{u} = 0, & \text{in } \Omega, \\ \mathbf{u} = 1\mathbf{i} + 0\mathbf{j}, & \text{on } \Gamma^{\text{up}} = \{0 \leq x \leq 1, y = 1\}, \\ \mathbf{u} = \mathbf{0}, & \text{on } \partial\Omega \setminus \Gamma^{\text{up}}. \end{cases}$$

- 1 Write the FEM for the problem, using the penalty method (1) for the imposition of Dirichlet BCs.
- 2 Write a function that assembles (separately) the matrices $A, -B, B^T, \mathbf{F}$ defining the algebraic form of the FEM approximation of the problem of point 1, using the stable pair of spaces $\mathbb{P}^2/\mathbb{P}^1$.

$$\left(\begin{array}{c} \text{notation from monolithic problem} \end{array} \begin{bmatrix} A & B^T \\ -B & 0 \end{bmatrix} \begin{bmatrix} \mathbf{U} \\ \mathbf{P} \end{bmatrix} = \begin{bmatrix} \mathbf{F} \\ \mathbf{0} \end{bmatrix} \right)$$

- 3 By block Gaussian elimination, the previous system can be reduced to the equation

$$S\mathbf{P} = \boldsymbol{\chi}, \quad (2)$$

where $S = BA^{-1}B^T$ is the Schur complement and $\boldsymbol{\chi} = BA^{-1}\mathbf{F}$. Write a function that constructs the matrix S and the associated KSP solver. This function should also take a bilinear form as an input, used to create a preconditioner for S .

- 4 Solve problem (2) by the preconditioned conjugate gradient method, with the pressure mass matrix M_p as preconditioner.

- 5 Repeat now the previous step using, as preconditioner:

- the identity matrix (i.e. no preconditioner);
- the pressure mass matrix M_p ;
- the lumped pressure mass matrix: $(\tilde{M}_p)_{ij} = \begin{cases} \sum_{k=0}^{N_p} (M_p)_{ik} & \text{if } i = j, \\ 0 & \text{otherwise.} \end{cases}$
- the diagonal of the pressure mass matrix: $(D_p)_{ij} = \begin{cases} (M_p)_{ij} & \text{if } i = j, \\ 0 & \text{otherwise;} \end{cases}$

How many iterations are needed to reach convergence?

Exercise 2

Let us now consider the unsteady Stokes problem in the unit square $\Omega := (0, 1)^2$

$$\begin{cases} \frac{\partial \mathbf{u}}{\partial t} - \Delta \mathbf{u} + \nabla p = 0, & \text{in } \Omega \quad \forall t \in (0, 1), \\ \operatorname{div} \mathbf{u} = 0, & \text{in } \Omega \quad \forall t \in (0, 1), \\ \mathbf{u}(t, \mathbf{x}) = t\mathbf{i} + 0\mathbf{j}, & \text{on } \Gamma^{\text{up}} = \{0 \leq x \leq 1, y = 1\} \quad \forall t \in (0, 1), \\ \mathbf{u} = \mathbf{0}, & \text{on } \partial\Omega \setminus \Gamma^{\text{up}} \quad \forall t \in (0, 1), \\ \mathbf{u} = \mathbf{0}, & \text{in } \Omega \quad \text{for } t = 0. \end{cases}$$

Solve the problem using the building blocks created in Exercise 1.